College Football Bowl Game Predictor

Machine Learning Mini-Project — UE24CS352A

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1. Problem Statement

The objective is to predict college football bowl-game outcomes using teams' regular-season performance. The model estimates a likely winner from statistical patterns in prior games.

2. Reference and Dataset

This project is based on the supplied Stanford CS229 report, *College Football Bowl Game Predictor*. The original study used regular-season and bowl-game statistics from 2000–2010, approximately 350 bowl games and 36 features per team. It compared score regression with score-differential regression.

This implementation retrieves completed games from ESPN's public college-football scoreboard endpoint for configured seasons. It derives average points scored, points allowed, points-scored variability, win rate and games played. This is not the same dataset as the original Stanford study; exact coverage is recorded after the pipeline runs.

3. Methodology

Score regression: two linear-regression models predict the points scored by each team, and the larger prediction determines the winner. **Differential regression:** a linear-regression model directly predicts Team A's score minus Team B's score; the sign determines the selected winner.

Features are calculated from regular-season games completed before each bowl game. Evaluation uses leave-one-season-out testing: the test season is excluded from model training. Metrics include winner accuracy and mean absolute error.

4. Implementation

Python is used with pandas for data preparation, scikit-learn for regression and evaluation, requests for data retrieval, matplotlib for a comparison chart and Streamlit for a dashboard. The code stores downloaded data, processed features, predictions and metrics in separate folders.

5. Results and Conclusion

Complete this section after running the project. Copy the actual accuracy and error metrics from `outputs/metrics.json` and include the generated model-comparison chart. The Stanford report's reported 66% accuracy is a reference result and must not be presented as this implementation's result.

The implementation follows the reference's central comparison but uses a smaller set of score-derived features. Future improvements include richer team statistics, feature selection and explicit upset prediction.